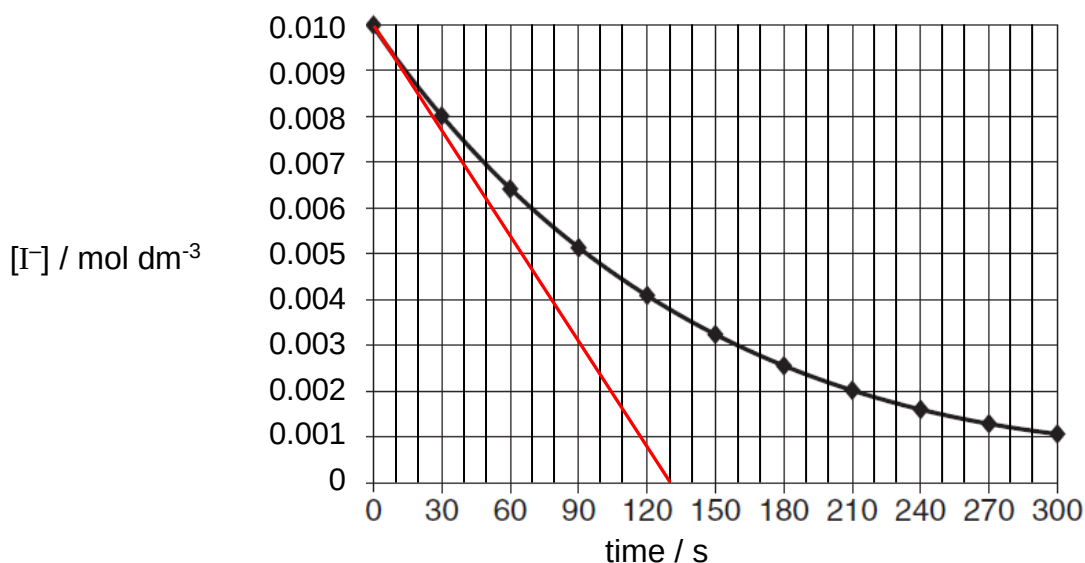


**MJC Prelim 2012 H1 Chemistry (Suggested Answers)****1 (a)**

$$\text{rate} = \frac{0.010}{130} = 7.69 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$$

(b) Half-life is constant at 90 s.**(c) Rate $\propto 1/t$**

Comparing experiment 2 and 3, when $[\text{H}^+]$ increased 5 times, rate remains constant.

Order of reaction with respect to $\text{H}^+ = 0$.

Comparing experiment 1 and 2,

$$\frac{1/16}{1/40} = \frac{(0.050)^x \times (0.010) \times (0.050)^0}{(0.010)^x \times (0.020) \times (0.050)^0}$$
$$x = 1$$

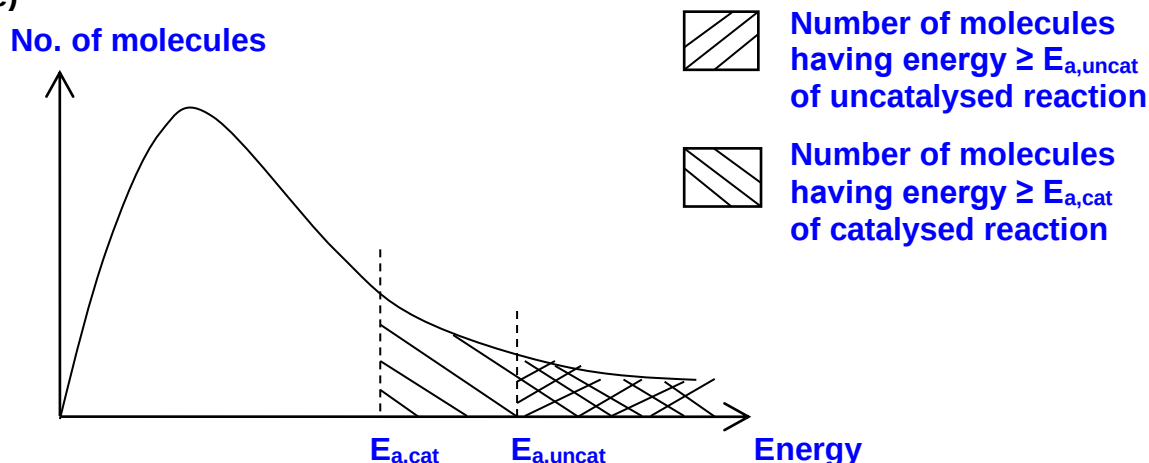
Order of reaction with respect to $\text{H}_2\text{O}_2 = 1$.

(d) (i) rate = $k [\text{H}_2\text{O}_2] [\text{I}^-]$

(ii) $7.69 \times 10^{-5} = k \times (0.050) \times (0.010)$
 $k = 0.154 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$



(e)



MnO_2 catalyst can increase the rate of reaction by providing an alternative reaction pathway of lower activation energy.

Fraction of reacting H_2O_2 particles with energy $\geq E_{a,\text{cat}}$ increased. Frequency of effective collisions increased. Since rate is proportional to the frequency of effective collisions, rate of H_2O_2 decomposition increased.

- (f) (i) Number of mole of $\text{Na}_2\text{S}_2\text{O}_3$
 $= 20 / 1000 \times 0.10 = 2.00 \times 10^{-3} \text{ mol}$

Number of mole of I_2
 $= 1.00 \times 10^{-3} \text{ mol}$

- (ii) Number of mole of H_2O_2 left from Reaction I
 $= 1.00 \times 10^{-3} \text{ mol}$

Number of mole of H_2O_2 added initially
 $= 5.00 \times 10^{-3} \text{ mol}$

Number of mole of H_2O_2 reacted in Reaction I
 $= 4.00 \times 10^{-3} \text{ mol}$

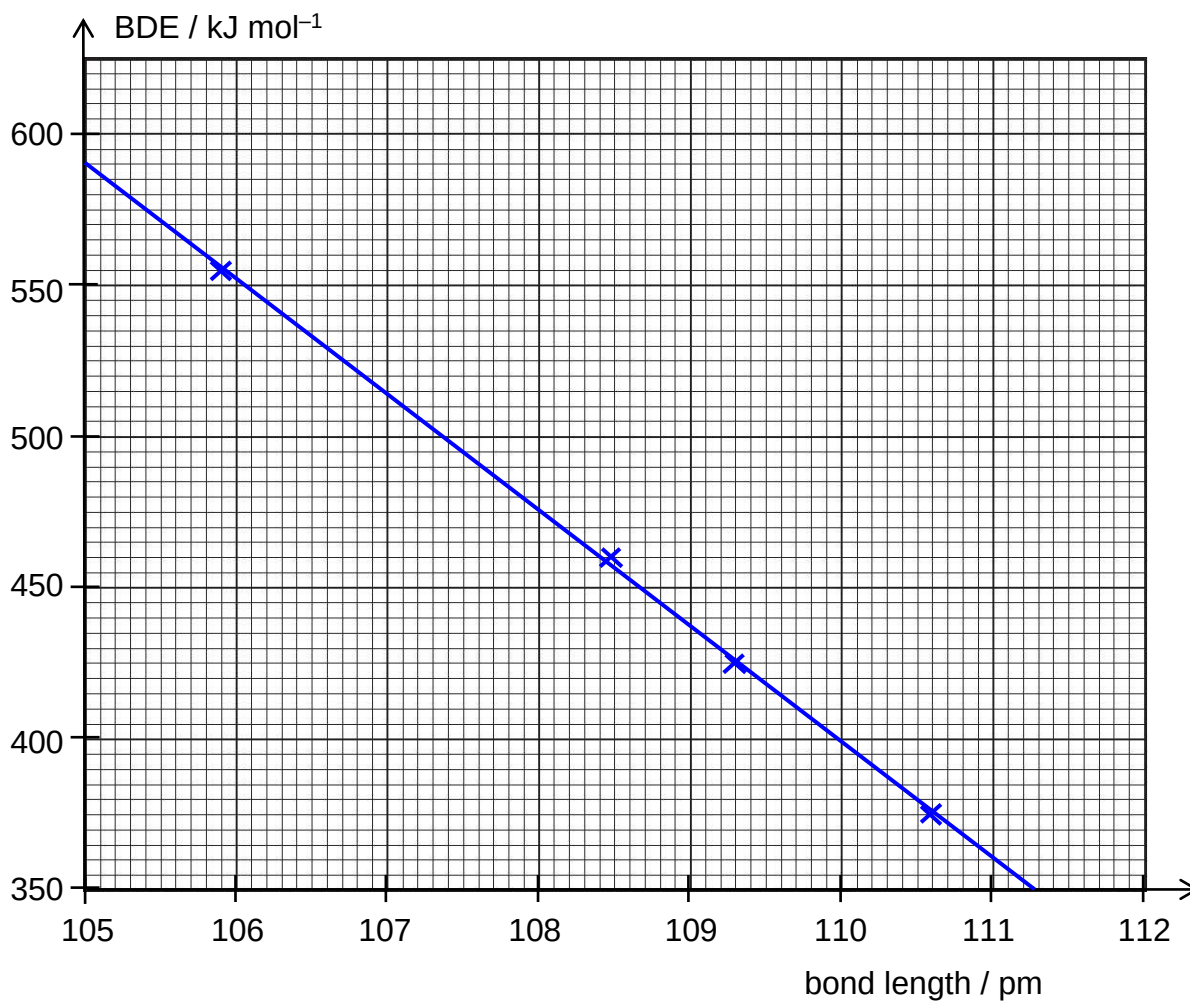
Number of mole of NaBr reacted in Reaction I
 $= 40 / 1000 \times 0.10 = 4.00 \times 10^{-3} \text{ mol}$

Hence, $4.00 \times 10^{-3} \text{ mol } \text{H}_2\text{O}_2 \equiv 4.00 \times 10^{-3} \text{ mol } \text{Br}^-$.

1 mole of H_2O_2 reacts with 1 mol of Br^- .



2 (a) (i)



(ii) The greater the extent of orbitals overlap, the shorter the bond length and the higher the BDE.

(b) (i) $\Delta H = -14 + (-36) - 12 = -62 \text{ kJ mol}^{-1}$

(ii) $-62 = \text{BDE}(\text{C-H}) + 193 - 244 - 366$
 $\text{BDE}(\text{C-H}) = 355 \text{ kJ mol}^{-1}$

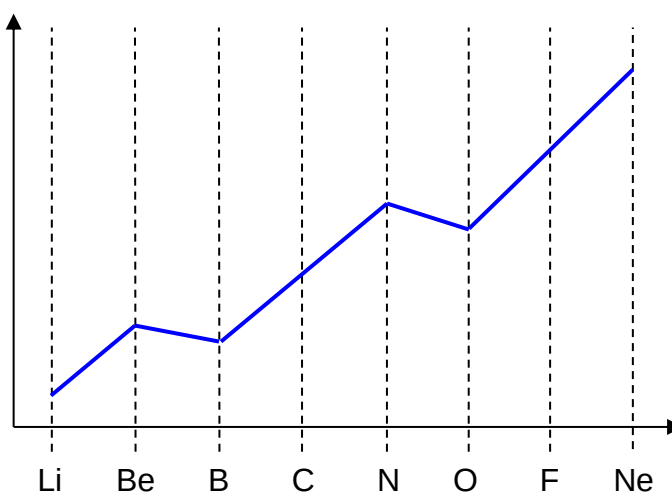
(iii) bond length is 111.2 pm.

(iv) Bond energy and BDE are for bonds in gaseous molecules. Some reactants and products in the reaction are liquids.

(c) (i) The C–O bond is more polar than the C–N bond.

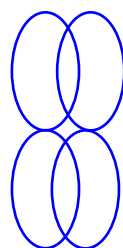


(ii)

first ionisation
energy

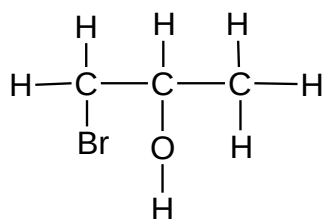
(iii) There is inter-electron repulsion of paired 2p electrons in the doubly-filled 2p orbital of O

(d) (i)



p-orbitals overlap sideways

(ii)





3 (a) $\text{H}_2\text{O} \equiv 2\text{H}$

$$\text{Mass of H in sample} = \frac{0.155}{18} \times 2 \times 1.0 = 1.72 \times 10^{-2} \text{ g}$$

$$\text{Percentage of H in sample} = \frac{1.72 \times 10^{-2}}{0.200} \times 100 = 8.6\%$$

$$\begin{aligned} \text{\% of Cl in sample} &= 100 - 57.4 - 8.6 \\ &= 34.9\% \end{aligned}$$

Empirical formula of compound A = $\text{C}_5\text{H}_9\text{Cl}$

(b) A undergoes oxidation with KMnO_4 to give colourless gas.

- A has terminal alkene group.

(c) Type of reaction for A to B : addition

Type of reaction for B to C: substitution

(d) C undergoes oxidation with alkaline aqueous iodine to give yellow solid.

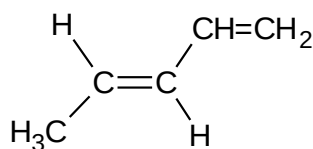
- One C has two $\text{CH}_3\text{CH}(\text{OH})-$ groups.

(e) A: $\text{CH}_2=\text{CHCH}_2\text{CHC}/\text{CH}_3$

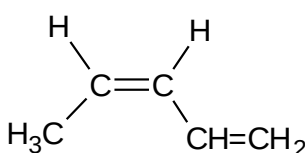
B: $\text{CH}_3\text{CHC}/\text{CH}_2\text{CHC}/\text{CH}_3$

C: $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$

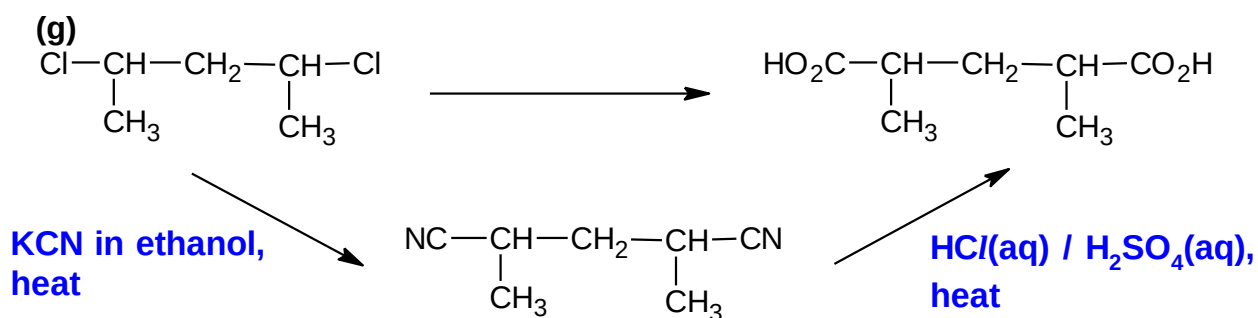
(f) Geometric / cis-trans isomerism.



trans-isomer



cis-isomer

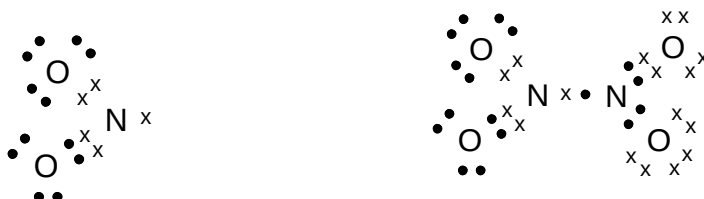




4 (a) (i)

$$K_c = \frac{[\text{N}_2\text{O}_4]}{[\text{NO}_2]^2} = \frac{\left(\frac{0.41}{0.28}\right)}{\left(\frac{0.18}{0.28}\right)^2} = 3.54 \text{ mol}^{-1} \text{ dm}^3$$

(ii)

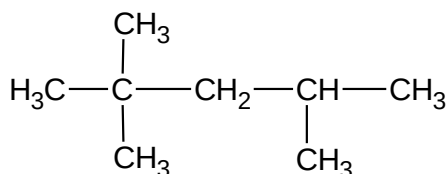


The reaction involves bond forming between two NO_2 to become one N_2O_4 . The reaction is exothermic.

(iii) The reaction above is exothermic.

Equilibrium position shifted left, towards the endothermic reaction to absorb added heat. More NO_2 produced / formed to give brownish colour of smog.

(b) (i)

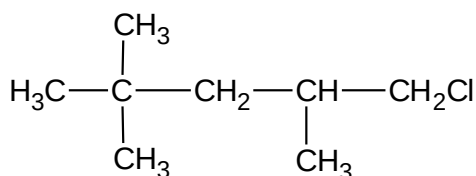
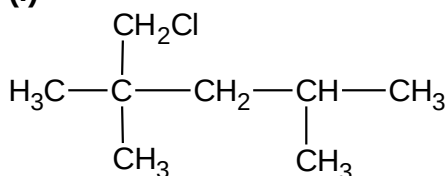
(ii) Heat absorbed by solution = $240 \times 4.2 \times 33 = 33264 \text{ J}$

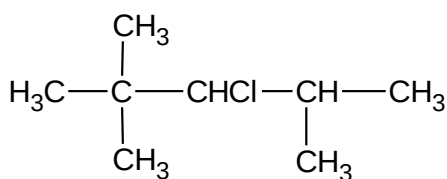
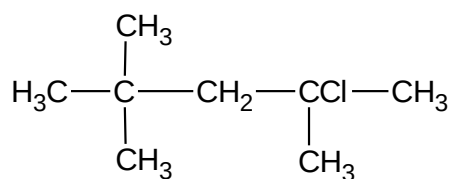
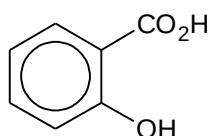
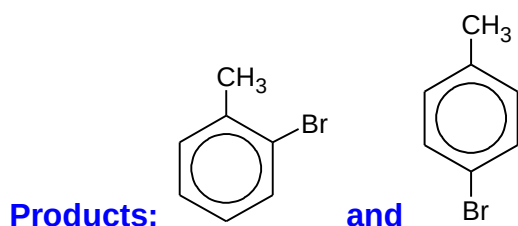
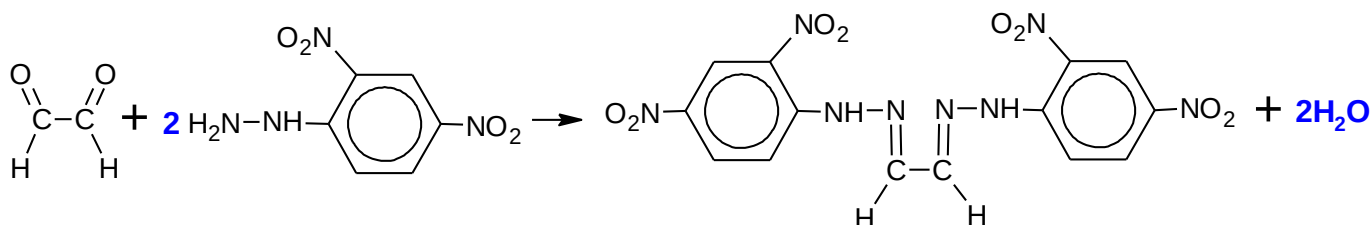
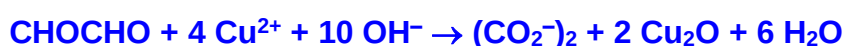
$$\text{Heat released by reaction} = 33264 \times \frac{100}{70} = 47520 \text{ J}$$

$$\text{Amount of 2,2,4-trimethylpentane burnt} = \frac{1}{114} = 8.772 \times 10^{-3} \text{ mol}$$

$$\begin{aligned} \text{Enthalpy change of combustion} &= -\frac{47520}{8.772 \times 10^{-3}} \\ &= -5417300 \text{ J mol}^{-1} = -5420 \text{ kJ mol}^{-1} \end{aligned}$$

(c) (i)



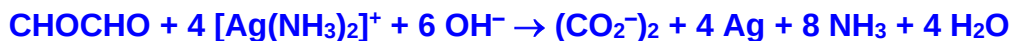
**A: primary****C: secondary****B: primary****D: tertiary****(ii) A : B: C: D = 9 : 6 : 4 : 3****(d)****(e) Conditions: uv light****Products: $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$ and $\text{C}_6\text{H}_5\text{CHBr}_2$** **Conditions: AlBr_3 / FeBr_3 / Fe** **5 (a) Primary alcohol and carboxylic acid****(b) Test: Add 2,4-DNPH to each compound separately and heat.****Observation: For ethanedial, orange ppt of hydrazone is observed.****For glycolic acid, no orange ppt is observed.****Test: Add Fehling's solution to each compound separately and heat.****Observation: For ethanedial, brick-red ppt of Cu_2O is observed.****For glycolic acid, no brick red ppt is observed.**



Test: Add Tollens' reagent to each compound separately and heat.

Observation: For ethanedial, silver mirror is observed.

For glycolic acid, no silver mirror is observed.



- (c) (i) A weak Bronsted acid is one which dissociates partially in solution to donate protons, H^+ .

$$(ii) K_a = \frac{[\text{HOCH}_2\text{CO}_2^-] [\text{H}^+]}{[\text{HOCH}_2\text{CO}_2\text{H}]}$$

- (iii) Since $[\text{CH}_2(\text{OH})\text{COO}^-] = [\text{H}^+]$,

$$K_a = \frac{[\text{H}^+]^2}{[\text{HOCH}_2\text{CO}_2\text{H}]}$$

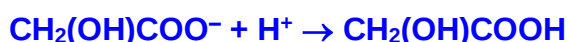
$$1.48 \times 10^{-4} = \frac{[\text{H}^+]^2}{0.20}$$

$$[\text{H}^+] = 5.44 \times 10^{-3} \text{ mol dm}^{-3}$$

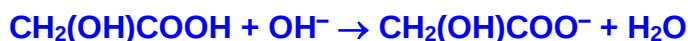
$$\text{pH} = -\lg (5.44 \times 10^{-3}) = 2.26$$

- (iv) A buffer solution is one which is capable of maintaining a fairly constant pH (by resisting pH change) when small amounts of acid or base are added to it.

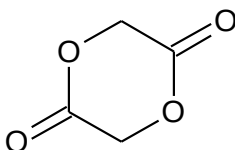
When small amount of H^+ is added,



When small amount of OH^- is added,

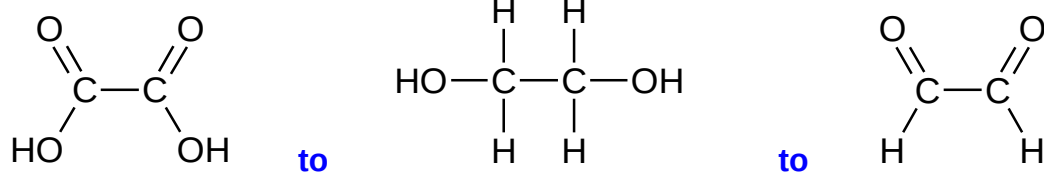


(d)





(e) (i)

Step I: LiAlH₄ in dry ether, r.t.p.Step II: K₂Cr₂O₇(aq), H₂SO₄(aq), heat under distillation

(ii) The intermolecular forces of attraction between ethanedioic acid molecules is stronger hydrogen bond, while weaker van der Waals' forces of attraction exists between ethanedial molecules.

6 (a) (i) CH₂(OH)CH(OH)CH₂OH(ii) CH₂=CHCOOH(b) (i) CH₂=CHCH₂Cl(ii) React with dilute H₂SO₄(aq) / NaOH(aq), heat(iii) CH₂=CHCHO(iv) SeO₂ + 2OH⁻ → SeO₃²⁻ + H₂O

SeO₂ is an acidic covalent oxide and hence would be expected to react with a base.

(c) (i) Structural (functional group) isomerism

(ii) Disproportionation has occurred simultaneously.

(d) (i) 1 CH₂=CHCH₂O⁻Na⁺2 CH₂Cl/CHCl/CH₂OH

(ii) Mg has higher electrical conductivity than Na.

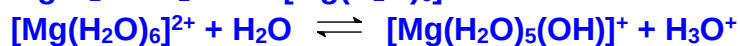
Mg has more delocalised mobile electrons as charge carriers.

(iii) Magnesium Chloride:

hydrolysed partially

pH 6.5

universal indicator: yellowish-green / greenish-yellow



Silicon Chloride:

hydrolysed completely

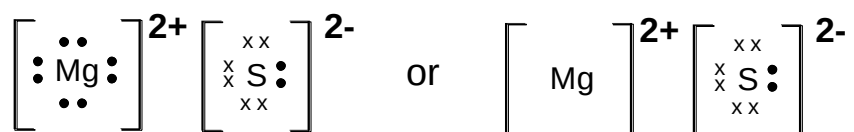
gives white fume of HCl

universal indicator: red

pH 2



(iv) 1



2 Not a redox reaction, no change in oxidation state.